MeshLink Server: Administrator Manual

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MeshLink is a self-hosted service mesh control plane designed for hybrid Kubernetes and bare-metal deployments. This manual covers installation, configuration, operational procedures, and troubleshooting for system administrators. The guidance in this document applies to MeshLink Server release 4.8.x; consult the migration notes when upgrading from 3.x or earlier.

1. System Requirements

1.1 Minimum Hardware

MeshLink Server requires a minimum of 4 CPU cores, 8 GB of RAM, and 50 GB of available disk space on the control plane host. For production deployments serving more than 500 mesh nodes, the recommended configuration increases to 8 CPU cores, 16 GB of RAM, and 200 GB of disk space provisioned on SSD storage. Network connectivity must support a sustained throughput of at least 100 Mbps with latency below 50 ms between the control plane and managed nodes.

1.2 Software Dependencies

The host operating system must be Ubuntu 22.04 LTS, Ubuntu 24.04 LTS, or Red Hat Enterprise Linux 9.x. Container runtime support is provided through Docker 24.0 or later, or containerd 1.7 or later. Kubernetes integration requires cluster version 1.28 or higher. PostgreSQL 14 or later is required as the persistent backing store; SQLite is supported only for single-node evaluation deployments and must not be used in production.

1.3 Network Ports

MeshLink Server uses three network ports by default. Port 8443 serves the administrative HTTPS API and the web console. Port 9100 exposes the Prometheus metrics endpoint and must be reachable only from the monitoring subnet. Port 7946 is used for inter-node gossip on both TCP and UDP and must be open between all control plane replicas in a high-availability deployment. All three ports may be remapped through the configuration file described in section 2.

2. Installation

2.1 Package Installation

MeshLink Server is distributed as a Debian package, an RPM package, and a container image. To install on Ubuntu, download the package from the official repository and install it using apt:

```
curl -fsSL https://repo.meshlink.example/install.sh | sudo bash
sudo apt update && sudo apt install -y meshlink-server
```

The installer creates a dedicated service account named meshlink under which the server process runs. The default configuration file is created at /etc/meshlink/server.yaml, and persistent state is stored under /var/lib/meshlink. Log files are written to /var/log/meshlink/server.log with automatic rotation at 100 MB per file, retaining the most recent 10 archives.

2.2 Initial Configuration

Before starting the service for the first time, the administrator must configure the database connection string, the cluster bootstrap token, and the TLS certificate paths. The bootstrap token must be a cryptographically random value of at least 32 bytes, base64 encoded. It is used exclusively during the cluster formation phase and must be rotated within 24 hours of the initial cluster join.

2.3 Database Migration

On first start, MeshLink Server applies all pending schema migrations automatically. When upgrading between minor versions, migrations are forward-compatible and may be applied while the service is running. Major version upgrades require the migration step to run with the service stopped; failure to observe this requirement may result in data corruption that requires restoration from backup.

3. Configuration Reference

3.1 Configuration File Format

The configuration file uses YAML syntax. The top-level keys are server, database, tls, telemetry, and cluster. Each key contains a nested object whose schema is described in the sections below. Configuration values may be overridden at runtime by environment variables prefixed with MESHLINK_, with nested keys joined by underscores. For example, the database host is overridden by MESHLINK_DATABASE_HOST.

3.2 Telemetry Configuration

Telemetry collection is enabled by default and sends anonymised usage statistics to telemetry.meshlink.example over HTTPS. The data collected includes the MeshLink version, the operating system family, the number of managed nodes, and aggregate request counts. No customer data, configuration values, or personally identifiable information is transmitted. To disable telemetry, set the value of telemetry.enabled to false in the configuration file or set the environment variable MESHLINK_TELEMETRY_ENABLED to false.

3.3 High Availability

MeshLink Server supports active-active high availability through a Raft consensus cluster of three or five nodes. The minimum cluster size for production is three nodes; clusters of two nodes are not supported because they cannot maintain quorum in the event of a single node failure. Cluster nodes elect a leader using Raft, and write operations are forwarded to the leader transparently. Read operations may be served by any node, with eventual consistency guarantees within a typical replication lag of under 200 milliseconds.

4. Operations

4.1 Service Management

MeshLink Server is managed through systemd on Linux hosts. The service unit is named meshlink-server.service. Standard systemctl commands apply for start, stop, restart, and status operations. The service exposes a graceful shutdown mechanism: on receipt of SIGTERM, the server completes any in-flight requests, drains the connection pool, and writes a final state checkpoint before exiting. The default shutdown timeout is 30 seconds, after which the process is forcibly terminated.

4.2 Backup and Restore

Backups consist of two artefacts: the PostgreSQL database dump and the contents of the /var/lib/meshlink directory. Both must be captured at the same point in time for the restore to succeed. The bundled meshlink-backup utility coordinates this by quiescing the server, capturing both artefacts atomically, and resuming operations. Backups are written to the path specified by the backup.destination configuration key, which supports local file paths, S3-compatible URLs, and SFTP endpoints. Backups are encrypted with AES-256-GCM using a key derived from the configured backup passphrase.

4.3 Log Rotation and Audit

Standard logs rotate at 100 MB as noted in section 2.1. Audit logs, which capture all administrative actions, are stored separately at /var/log/meshlink/audit.log and are not rotated automatically. The administrator is responsible for archiving audit logs to long-term storage. The audit log records the actor identity, source IP address, action verb, target resource, and outcome for every administrative operation, formatted as one JSON object per line.

5. Security

5.1 Authentication

MeshLink Server supports three authentication methods: local username and password, OpenID Connect, and mTLS client certificates. Local authentication uses Argon2id for password hashing with the default parameters of 64 MB memory cost, 3 iterations, and 4 parallel lanes. OpenID Connect requires the issuer URL, client ID, and client secret to be set in the auth.oidc configuration block. Multi-factor authentication is supported through TOTP and may be enforced organisation-wide.

5.2 Role-Based Access Control

Authorisation is enforced through a built-in RBAC system. Four roles are provided out of the box: Viewer (read-only access), Operator (read and operational actions, no configuration changes), Administrator (full access except user management), and Owner (full access including user and role management). Custom roles may be defined by combining the underlying permission verbs documented in appendix A.

5.3 Certificate Management

Server certificates may be provided manually or auto-renewed through ACME. When ACME is enabled, MeshLink Server obtains certificates from the configured directory URL, with Let's Encrypt as the default. Certificates are renewed automatically when fewer than 30 days remain before expiry. The renewal process is logged to the standard log file with the prefix [acme]. If renewal fails, an alert is raised through the configured notification channel.

6. Troubleshooting

6.1 Common Issues

Service fails to start: check the configuration file syntax with meshlink-server validate-config. Verify the database is reachable and the credentials in the configuration file are correct. Confirm the listening ports are not occupied by another process.

High memory usage: the most common cause is unbounded query result sets. Set the server.max_response_rows configuration value to limit the maximum rows returned by API responses; the default is 10000.

Cluster split-brain: indicates a network partition. Restore connectivity between all cluster members and use meshlink-cluster heal to reconcile the state. Do not manually edit the Raft log; this will cause permanent data loss.

6.2 Diagnostic Bundle

When opening a support case, attach a diagnostic bundle generated by running meshlink-server diagnostics --output bundle.tar.gz. The bundle contains configuration files (with secrets redacted), the most recent 24 hours of logs, the current cluster topology, and database schema version information. The bundle does not include audit logs or backup artefacts.

7. Quick Reference

Setting	Default Value
Admin API port	8443
Metrics port	9100
Gossip port	7946 (TCP/UDP)
Config file path	/etc/meshlink/server.yaml
State directory	/var/lib/meshlink
Standard log path	/var/log/meshlink/server.log
Audit log path	/var/log/meshlink/audit.log
Log rotation size	100 MB
Log archive retention	10 files
Min HA cluster size	3 nodes
Replication lag (typical)	under 200 ms

Shutdown timeout	30 seconds
Cert renewal threshold	30 days before expiry
Default max response rows	10,000
Password hash	Argon2id (64MB, 3 iter, 4 lanes)